Data Analysis Pipeline

Chapter 1

Ashim Khadka

Outline

1. Knowledge Discovery from Database Process

What is Data?

- Imagine you need a ride to the cinema:
  - You open a ride-sharing app on your phone.
  - You enter the cinema's location.
  - The phone automatically detects your current location.
  - The app shows ride options with different prices.
  - You select one and confirm the booking.
  - You receive:
    - Driver's name
    - License plate number
    - Vehicle make and model
    - Driver's photo
  - A real-time map shows the driver's location and estimated arrival time.

Why Data Matters

- **We constantly use data to make decisions**, often without even realizing it.
- The way data is **presented** directly affects how easy it is to understand and use.
- In ride-sharing, we:
  - Enter data (destination)
  - Receive data (options, prices)
  - Use this data to make a decision
- Additional data like:
  - Driver and vehicle details
  - Real-time tracking on a map

  help us feel informed and secure.
- **Conclusion:** Effective data presentation enhances understanding and supports better decisions.

2. Data

Data

Philosophical Context:

- plural of datum, Latin for "something given") refers to known or assumed facts
- data are considered foundational truths or starting points for reasoning

Computing Context:

- Data are sequences of bits stored on digital devices
- may be structured (e.g., database records), semi-structured (e.g., XML, JSON), or unstructured (e.g., images, videos, audio)

Data refers to raw facts, observations, or measurements that are collected, stored, and processed for analysis to generate useful insights

Characteristics of Data

1. **Raw and Unprocessed**
   - has not yet been cleaned, categorized, or interpreted: **raw input**
   - before analysis, data is just a collection of facts without meaning
   - e.g: When the result of a particular test is declared it contains **data** of all students
   - after analysis, it provides meaning
     - When you find the marks you have scored you have the **information** that lets you know whether you have **passed** or **failed**
2. **Context-dependent**
   - same data can mean different things depending on the situation
   - without knowing the context, you cannot extract meaningful insights
3. **Can be in any form**
   - Data is not just numbers or text. It can be visual, audio, video, or even sensor readings

Metadata

- data about data
- provides context and meaning
- Given data: [69, 72, 73, 64, 66]
    - Without metadata: meaningless
    - With metadata:
        - If heights in inches $\rightarrow$ insights about physical characteristics
        - If votes in an election $\rightarrow$ insights about political outcomes
- Metadata enables **interpretation**
    - without it, data has no meaning and cannot yield knowledge

Example: Counting Vehicles

- Suppose you want to find out how many vehicles pass a certain point on a road in a 10-minute interval.
  - You stand near the road and count each vehicle that passes by during that 10-minute window.
- To get more comprehensive data:
  - Repeat the counting during different times of the day (morning, afternoon, evening).
  - Perform the counting on different days (weekdays, weekends).
- This process gives you multiple observations, which can then be analyzed to find:
  - Traffic patterns
  - Peak traffic hours
  - Differences between weekdays and weekends

Data Analysis

Based on the data collected (e.g., from counting vehicles), we can perform various types of analysis:

- How many **public** and **private** vehicles passed during each interval?
- What is the count of different types of vehicles, such as:
  - Cars
  - Taxis
  - Motorcycles
  - Buses and trucks
- How do these counts vary:
  - At different times of the day?
  - On weekdays vs. weekends?
- $\cdots$ and many other insights depending on how the data is categorized and analyzed.

Data Analysis

Systematic computational process of examining data to:

- discover useful patterns
- extract meaningful information
- support decision-making

To perform analysis:

- data must be clean, structured, and relevant
- requires understanding data types, variables, and relationships

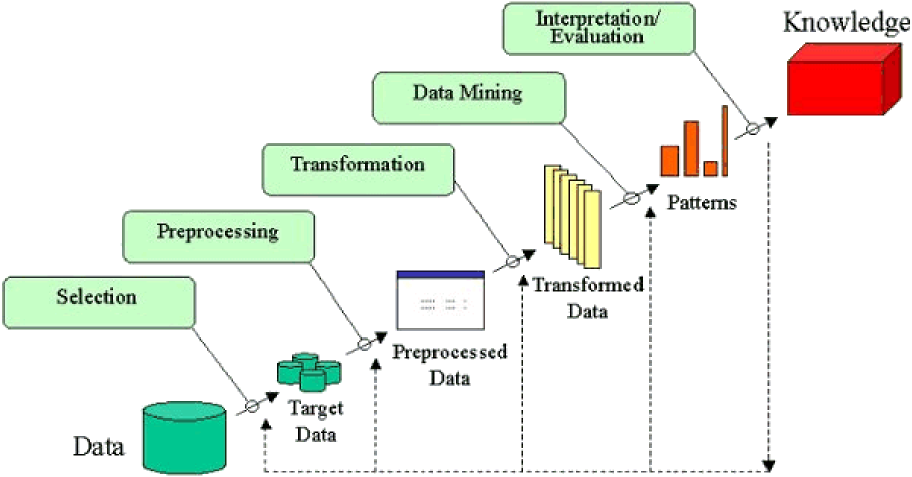
Data becomes useful only when:

- helps answer specific questions or solve problems
- computational process uses statistical, mathematical, or machine learning methods to extract information and knowledge from data.

Knowledge Discovery from Data (KDD)

- **data rich, but information poor**
- massive amounts of data from sources like business transactions, scientific data, and the web
  - lack the tools to extract meaningful knowledge from it
- KDD: extraction of interesting patterns or knowledge from huge amounts of data

KDD


Interpretation/
Evaluation
Knowledge
Data Mining
Transformation
Preprocessing
Selection
Patterns
Transformed
Data
Preprocessed
Data
Target
Data
Data

Data Selection

Data Selection is the initial step in the KDD process, where relevant data is identified and chosen for analysis.

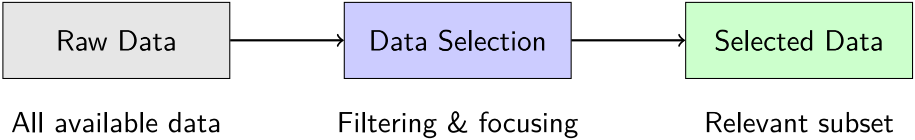
- Selecting a dataset or focusing on specific variables
- Choosing samples or subsets of data
- Narrowing down based on task goals

Why is Data Selection Important?

- Ensures only **most relevant data** is used
- Improves **efficiency** and **accuracy**
- Requires thorough understanding of the **application domain**

Outcome: Accurate, relevant, and actionable insights.

Data Selection: Visual Overview



Critical Point

Data selection must be guided by **domain knowledge** – understanding what variables actually matter for the analysis goal

Data Cleaning and Preprocessing

Data cleaning ensures the dataset is accurate and reliable by correcting errors, handling missing values, removing duplicates, and addressing noisy or outlier data

Three Main Techniques:

Issue	Solution
Missing Values	Fill with mean, median, or most probable value
Noisy Data	Smooth using binning, regression, or clustering
Duplicate Records	Remove to maintain consistency

Goal

Enhance data quality and improve the effectiveness of data mining

Data Cleaning: Before vs After

Before Cleaning:

- Missing values: ?, NULL, NA
- Outliers: 1000, -999
- Duplicates: Same row x3
- Inconsistent formats: "Jan", "January", "01"

After Cleaning:

- Missing values: Imputed (mean/median)
- Outliers: Capped or removed
- Duplicates: Unique rows only
- Consistent formats: Standardized

Clean data = Reliable insights

Data Transformation

converts data into a format more suitable for analysis

Techniques:

Technique	Description
Normalization	Scaling data to a common range (e.g., 0 to 1)
Discretization	Converting continuous data into discrete categories
Data Aggregation	Summarizing multiple points (averages, totals)
Concept Hierarchy	Organizing data into hierarchies (e.g., city $\rightarrow$ state $\rightarrow$ country)

Normalization

$$x_{\text{norm}} = \frac{x - \min(x)}{\max(x) - \min(x)}$$

Raw values [200, 700, 2333] $\rightarrow$ Normalized [0, 0.23, 1]

Data Reduction

simplifies the dataset while preserving key information

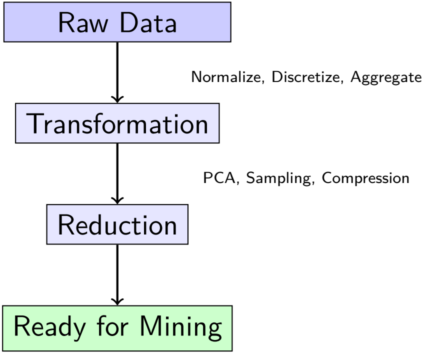
Three Main Approaches:

- **Dimensionality Reduction (e.g., PCA)**
  Reduces number of variables while keeping essential information
- **Numerosity Reduction**
  Reduces data points using sampling methods (random, stratified, cluster)
- **Data Compression**
  Compacts data for easier storage and processing

Goal

Make data **smaller but still representative** – faster processing without losing meaningful patterns.

Transformation & Reduction: Summary



Together They Ensure

Data is **ready for deeper analysis and mining**.

Data Mining

process of discovering valuable, previously unknown patterns from large datasets through automatic or semi-automatic means

Characteristics of Discovered Patterns:

Characteristic	Meaning
Validity	Patterns hold true even with new data
Novelty	Insights are non-obvious and surprising
Usefulness	Information can be acted upon
Understandability	Patterns are interpretable to humans

*"Data mining turns raw data into **actionable intelligence**."*

Choosing the Data Mining Task

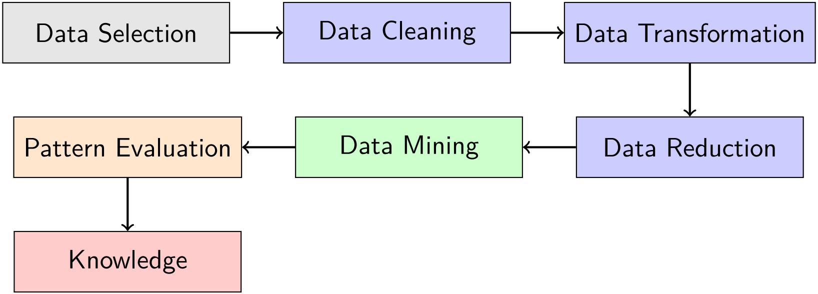
Common Tasks:

- **Classification**
  Predict a category (e.g., churn: Yes/No)
- **Regression**
  Predict a number (e.g., income $)
- **Clustering**
  Group similar items (e.g., customer segments)
- **Association**
  Find item relationships (e.g., market basket)

Example Algorithms:

- Decision Trees, SVM, k-NN
- Linear Regression, Random Forest
- k-Means, DBSCAN, Hierarchical
- Apriori, FP-Growth

The Complete KDD Process



Data flows from raw → prepared → mined → actionable insights

1. **Data Selection:** Choose relevant data guided by domain knowledge
2. **Data Cleaning:** Handle missing values, noise, and duplicates
3. **Data Transformation:** Normalize, discretize, aggregate, build hierarchies
4. **Data Reduction:** Use PCA, sampling, or compression to simplify
5. **Data Mining:** Discover valid, novel, useful, understandable patterns

Garbage In = Garbage Out (GIGO)

Quality preparation enables quality mining

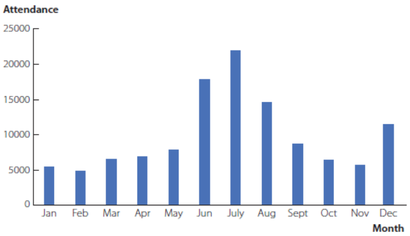
Example: Data visualization

DATA *file*

Zoo

TABLE 1.1	Zoo Attendance Data					
Month	Jan	Feb	Mar	Apr	May	Jun
Attendance	5422	4878	6586	6943	7876	17843
Month	July	Aug	Sept	Oct	Nov	Dec
Attendance	21967	14542	8751	6454	5677	11422

FIGURE 1.1 A Column Chart of Zoo Attendance by Month

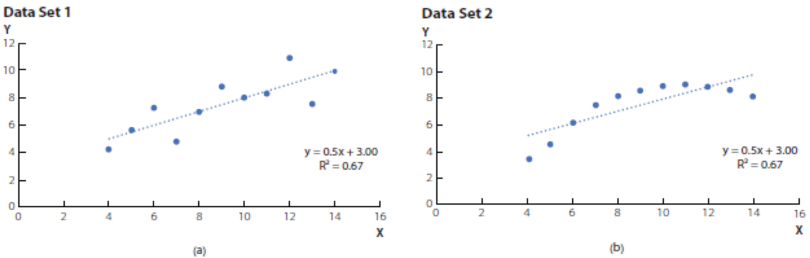


- What is the highest attendance month at the zoo?
- Why does the zoo attendance in December and January not follow these patterns?

Example: Data visualization

TABLE 1.2 Two Data Sets from Anscombe	Data Set 1		Data Set 2	
	X	Y	X	Y
	10	8.04	10	9.14
	8	6.95	8	8.14
	13	7.58	13	8.74
	9	8.81	9	8.77
	11	8.33	11	9.26
	14	9.96	14	8.1
	6	7.24	6	6.13
	4	4.26	4	3.10
	12	10.84	12	9.13
	7	4.82	7	7.26
	5	5.68	5	4.74
Average	9	7.501	9	7.501
Standard Deviation	3.317	2.032	3.317	2.032

FIGURE 1.2 Anscombe's Data Displayed Graphically



Summary

- Data: raw material. It needs context (metadata) to be useful
  - Data $\neq$ Knowledge
- must be interpreted via metadata and analyzed/visualized to become knowledge
- visualization and analysis are not competitors but collaborators
  - both aim at knowledge synthesis
- Both together make data science powerful:
  - Analysis tells you what the data says using computational methods
  - Visualization helps you and others understand it quickly

Data Types

Data Types

1. Dirty Data
2. Clean Data

Dirty Data

- contains errors, inconsistencies, or missing values: making it unsuitable for direct analysis
- Dirty Data:
  - Incomplete: Missing values or entire records
  - Inaccurate: Incorrect information, typos, or outdated data
  - Inconsistent: Data that doesn't follow the same format or rules
  - Duplicate: Redundant records or entries

Causes of Dirty Data:

- Human error during manual data entry (typos, misspellings)
- Issues during data migration or integration from different systems
- Lack of proper data validation checks during data collection

Name	Age	Email	Date	Purchase	Price
ram	28	ram@example.com	01/15/2024	Laptop	15000
hari		hari@example.com	2024-01-16	Mouse	Rs 1500
ram	28	ram@example.com	01/15/2024	Laptop	15000
raj	35	@example.com	12/15/2024	Laptop	Nrs15000

Why Cleaning Data is Important?

Dirty data can:

- Distort analysis
- Produce incorrect insights
- Mislead decision-making
- Cause errors in visualizations

Data cleaning is often the most time-consuming part of data analysis

- up to 80% of effort in real-world Project/research

Consequences of Dirty Data:

- Flawed insights and erroneous reports
- Poor decisions based on inaccurate information
- Wasted resources and increased operational costs
- Decreased efficiency and productivity

Clean Data

- free from errors, inconsistencies, and redundancies, making it suitable for analysis and decision-making

Characteristics of Clean Data:

- Accurate: Values are correct and reflect reality
- Complete: No missing values in important fields
- Consistent: Data is uniformly represented across all sources and records
- Unique: No duplicate entries for the same record
- Timely/Current: Data is up-to-date and relevant
- Valid: Data adheres to predefined rules and formats

Importance of Clean Data

- Enables accurate analysis and reliable insights
- Supports informed and effective decision-making
- Improves operational efficiency and reduces costs
- Ensures compliance with data protection regulations
- Enhances the performance of machine learning models and AI applications

data cleaning or data cleansing: process of transforming dirty data into clean data

- involves identifying, rectifying, or removing corrupt, inaccurate, or irrelevant records from a dataset to improve its overall quality

Name	Age	Email	Date	Purchase	Price
ram	28	ram@example.com	01/15/2024	Laptop	15000
hari	50	hari@example.com	01/16/2024	Mouse	1500
raj	35	raj@example.com	12/15/2024	Laptop	15000

- Data exists in many forms, and its classification often depends:
  - how organized and formatted it is
- leads to the fundamental distinctions
  1. structured
  2. semi-structured
  3. unstructured data

Structured Data

- highly organized and adheres to a predefined schema or model
- typically stored in tabular formats, like relational databases
  - data elements are clearly defined and relationships between them are explicitly established
- includes numbers (age, income, num.vehicles), text (housing.type), Boolean type (is.employed), and categorical data (gender, marital.stat)
- often used for transactional, operational, and analytical purposes

Characteristics

1. **Predefined Schema (Rigidity):** Before any data is entered, the "blueprint" must be created
   - must define what each column represents (e.g., Name, Date, Amount) and what kind of data is allowed there
2. **High Organization:** Data is mapped into fixed fields
   - Data elements are categorized and labeled, enabling systematic access and analysis
3. **Easy to Query:** Can be efficiently searched and manipulated using query languages like SQL
4. **Consistent Data Types:** Each field contains data of a specific type
   - integer, string, date, boolean

Age	Income	Housing	Employed	Gender	Marital
28	$55,000	Apartment	True	Female	Single
45	$110,000	Detached Home	True	Male	Married
34	$72,500	Townhouse	True	Female	Divorced
67	$48,000	Detached Home	False	Male	Widowed

Semi-structured Data

- doesn't conform to the rigid, tabular structure of relational databases
- does contain tags or markers that organize the data and establish a hierarchical structure
- Doesn't fit neatly into rows and columns but can still be parsed and analyzed programmatically
- **Examples:** XML (Extensible Markup Language), JSON (JavaScript Object Notation), CSV (Comma-Separated Values)

Characteristics:

1 **Flexible Schema:** Unlike structured data, semi-structured data allows for variations
   - one record might have five fields while the next has seven
2 **Self-Describing:** data contains its own metadata
   - labels (tags or keys) are stored alongside the value (e.g., <name>Alice</name>)
3 **Hierarchical Structure:** often uses a "parent-child" relationship (nested data) to organize information
4 **Programmatically Parsable:** humans can read it
   - specifically designed to be easily processed by machines and scripts

```
{
  "employee": {
    "id": 101,
    "name": "Alice Chen",
    "department": "Engineering",
    "skills": ["Python", "SQL", "Cloud"],
    "contact": {
      "email": "alice@company.com",
      "phone": "+1-555-0123"
    }
  }
}
```

Unstructured Data

- Unstructured data has no predefined format or organization
- does not fit into traditional row-column databases and typically lacks a fixed schema
  - text-heavy, multimedia-rich, or highly complex
- type of data accounts for the vast majority of data generated today
- **Examples:** Text Documents, Social Media Content, Multimedia Files
- roughly $80 - 90\%$ of all data generated in the digital world

Characteristics

1. **No Fixed Schema:** No rows/columns or predefined fields
2. **Native Format Storage:** Stored in original form (text, binary, streams)
   - `.wav` audio file or a `.jpg` image
3. **High Volume & Variety:** Represents $80 - 90\%$ of enterprise data
   - Demands scalable storage (data lakes, cloud)
4. **Qualitative & Subjective:** Often reflects opinions, emotions, experiences rather than quantitative
   - Analysis may involve sentiment, theme, or pattern detection
5. **Analysis Challenges:** difficult to search or categorize using traditional SQL-based tools
   - Needs semantic search, embeddings, or vector databases
6. **Time-Sensitive Value:** Real-time streams (e.g., social media) lose value quickly
   - Requires streaming analytics & low-latency processing
7. **Rich in Insights:** Contains nuanced human behavior & trends
   - High potential for competitive intelligence & innovation

Example

In an email, the sender, recipient, and timestamp are **structured data**, while the body of the email is **unstructured data**

Data sources

Data sources

Types of Data based on Originality

1. Primary Data Sources
   - Data collected directly by the researcher for a specific purpose
   - Original and unpublished data
   - **Advantages:** Tailored to specific needs, better control over data quality, up-to-date
   - **Disadvantages:** Time-consuming, costly, may require specialized tools or skills
   - **Examples:**
     - Customer satisfaction survey conducted by a company
     - Sensor data from IoT devices in a factory

2. Secondary Data Sources

- Data collected and published by others for a different purpose
- Readily available from various sources
- **Advantages:** Cost-effective, easy to access, useful for general context
- **Disadvantages:** May not be fully relevant, possibly outdated, unknown data quality
- **Examples:**
  - Government census reports
  - Published research articles or open data repositories

Data acquisition

Data acquisition

- process of collecting, measuring, and converting real-world information into digital data that can be used for analysis, visualization, and decision-making
- crucial first step in any data pipeline or analytics project
  - without data, there's nothing to analyze
  - helps collect up-to-date data (e.g., sensors in IoT devices) for real-time decision-making
- reliable and accurate data improves the quality of insights, leading to better business or research decisions

Methods of Data Acquisition

1. **APIs (Application Programming Interfaces)**
   - a set of rules that allows access to data or functionality from another
   - used to collect structured, real-time data from online platforms or services
   - requires API key or authentication
   - **Example:** Weather API, Twitter API
2. **Web Scraping**
   - Web scraping involves extracting data from websites by parsing HTML content
   - used when a website does not offer an API but the information is visible on the site
   - Access data from almost any website but can break if website structure changes
   - **Example:** News headlines, product prices

3. Sensors

- Sensors are physical devices that measure real-world phenomena and generate digital data
- used in IoT, smart devices, and industrial automation
- Real-time, continuous data collection from physical devices
  - can be noisy, needs cleaning
- **Example:** IoT temperature, heartbeat monitor

3. Data Preprocessing

Data Preprocessing

- Critical step in the data science pipeline
- Most available data is **messy** and **low quality**
- real-world data is often “dirty”-
  - containing errors, inconsistencies, and missing values
  - must be refined before it can be used for modeling
- process of transforming raw, messy, and unstructured data into a clean, structured format suitable for analysis, visualization
  - to generate insight and meaning

preprocessing aims to improve data quality across several dimensions

1. **Accuracy:** Ensuring data is free from errors and provides correct values
2. **Completeness:** Addressing data that lacks specific attributes of interest or contains only aggregate data
3. **Consistency:** Resolving discrepancies in naming conventions or data formats
4. **Timeliness, Believability, and Interpretability:** Ensuring the data is up-to-date, trusted, and easy for humans to understand

Three common tasks:

1. Data cleaning
2. Data integration
3. Data transformation and discretization

Data Cleaning

Data Cleaning

- Fix errors, handle missing values, and ensure consistency so the data is accurate and usable
- process of fixing or removing “noise” and inconsistencies
- often the most time-consuming part of preprocessing

Common actions

- Removing duplicates, Handling missing values (e.g., filling, dropping)
- Correcting typos or inconsistent spellings
- Standardizing date/time formats
- Filtering out irrelevant records

Handling Missing Data

Handling Missing Data

- Missing data occurs commonly in many data analysis applications
  - due to equipment malfunctions, user omissions, data corruption, or differences in data collection policies
- Missing data can lead to incorrect analysis if not handled properly
- uses the floating-point value `NaN` (Not a Number) to represent missing data

Techniques

1. Deletion Methods
2. Imputation Methods
   a. Statistical Substitution
   b. Predictive Imputation
   c. Multiple Imputation

Deletion Methods

Listwise Deletion: "All or Nothing" Approach

- If a single attribute is missing, the entire row (record) is deleted
- Can drastically reduce dataset size, especially if missingness is widespread across many variables

Pairwise Deletion: "Use What You Have" Approach

- record is only excluded when the missing variable is required for a specific calculation
- e.g., calculating the correlation between two specific columns
- Benefit: Preserves more data than listwise deletion

Employee	**Age**	**Annual Bonus ($)**	**Years at Company**
A	25	5000	2
B	30		5
C		7000	8
D	45	12000	15

Case 1: Listwise Deletion

- delete Employee B (missing Bonus) and Employee C (missing Age)
- left with only 2 employees (A and D) for further analysis

Case 2: Pairwise Deletion calculate the correlation between

Age and **Years at Company**

- A: (25, 2) - Available
- B: (30, 5) - Available
- C: (Missing, 8) - Excluded
- D: (45, 15) - Available
- Result: use 3 records (A, B, D) to calculate correlation

Bonus and **Years at Company**

- A: (5000, 2) - Available
- B: (Missing, 5) - Excluded
- C: (7000, 8) - Available
- D: (12000, 15) - Available
- Result: use 3 records (A, C, D) to calculate correlation

Table: Comparison of Pairwise vs. Listwise Deletion

Feature	**Pairwise Deletion**	**Listwise Deletion**
Data Volume	**High:** Preserves as much data as possible	**Low:** Can result in a very small sample size
Statistical Power	**Better:** More data points $\rightarrow$ more reliable results	**Worse:** lose more data over a few missing cells
Consistency	**Risky:** analysis will be based on different groups of data (different N values)	**Consistent:** Every calculation is based on the exact same "clean" group

Imputation Methods

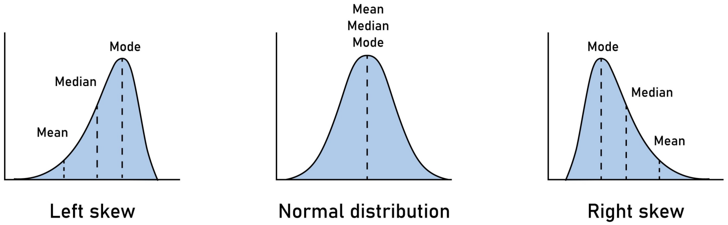
- statistical process of replacing missing data with substituted values
- *Rule:* Never impute your target variable
  - better to drop entire row or tuple
- Deleting data: often wasteful and can introduce bias if the missing values aren't random
- allows to keep sample size large while maintaining the statistical integrity of the dataset

Simple Approach:

- fills missing values with a specific value or method
  - forward fill: from previous value
  - backward fill: from next value

Statistical Substitution

1. **Mean:** Filling missing values with the average of the column. Best for normally distributed numerical data
   - Best for normally distributed continuous data
   - Data: "Normal" (a bell curve) and has no extreme outliers
2. **Median:** Filling with the middle value
   - Best for skewed continuous data and less sensitive to outliers
3. **Mode:** Filling with the most frequent value
   - Used for categorical data



Example

Student	Exam Score
A	80
B	85
C	
D	90
E	

Mean Imputation

- Calculation: $(80 + 85 + 90)/3 = \mathbf{85}$
- Result: Student C and E both receive a score of 85

Median Imputation

- Calculation: Sort the scores $(80, 85, 90)$
  - middle value is **85**
- Result: Student C and E both receive a score of 85

Example

Student	Exam Score
A	80
B	85
C	
D	90
E	
F	10

Mean Imputation

- Calculation: $(80 + 85 + 90 + 10)/4 = \approx \mathbf{67}$
- Result: Student C and E both receive a score of 67

Median Imputation

- Calculation: Sort the scores $(10, 80, 85, 90)$
  - middle value is **82.5**
- Result: Student C and E both receive a score of 82.5

Example

Scenario: A weather station records the temperature every hour

- brief 3-minute Wi-Fi outage, two readings are missing
- Data: $20°C, 21°C, 22°C, \textbf{NaN}, 22°C, 21°C$
- ***Mean:*** Temperature is a continuous variable that usually follows a "Normal Distribution"
    - changes gradually and stays near an average
    - **no outlier** temperatures like $100°C$ or $-50°C$ to mess up the average
- Result: fill the gap with $21.2°C$

Example

Scenario: Analyzing the "Starting Salaries" of 100 graduates from a university to help a student negotiate a job offer

- Data: Most graduates earn between \$50k and \$70k, but one graduate is a pro developer earning \$10 million a year
- ***Median:*** average salary = \$150k, which is a lie for 99% of the students
    - Result: using the Median, middle person in the list (who earns \$60k)
    - fill any missing values with \$60k, which is a much more realistic expectation for a typical student

Example

Scenario: An online clothing store is analyzing "Customer Preference" for shirt sizes

- Some customers bought shirts but didn't fill out their "Preferred Size" in their profile
- Data: Small, Medium, Medium, Large, Medium, Small, Medium, [Missing]
- ***Mode:*** cannot calculate the "average" of a size Small and a size Large
  - (Small + Large)/2 makes no sense
- Result: look for the Mode (the most frequent value), which is Medium
  - assume the missing customer most likely wants a Medium
- Standard for Categorical Data (colors, sizes, locations)

Predictive Imputation

- predictive technique
- Instead of just looking at one column (like Mean or Median)
  - looks at the relationships between all attributes (features) to make a calculated guess

1. **Regression Imputation:**

- Treats the missing variable as a target and uses other variables as features to predict it via linear or logistic regression
  - a *Linear Regression:* Numerical Data
    - missing value is a continuous number (like weight, price, or temperature)
  - b *Logistic Regression:* Categorical Data
    - missing value is a category or a binary choice (Yes/No, Pass/Fail, Spam/Not Spam)
- Preserves relationships between variables
- Cons: Can "overfit" the data, making relationships look stronger than they actually are

Regression Imputation: Example

Scenario: Analyzing a dataset of Used Cars

- One car has its "Mileage" and "Age" listed, but the "Selling Price" is missing
- Relationship: as a car gets older and has more miles, its price goes down
- $\text{Price} = \beta_0 - (\beta_1 \times \text{Age}) - (\beta_2 \times \text{Mileage})$
  - plugs the missing car's Age (e.g., 5 years) and Mileage (e.g., 40k)
- **Linear regression:** Price $\rightarrow$ continuous numerical data
- Result: predicts the price is $\$18,500$ and fills the gap with that specific value

Regression Imputation: Example

Scenario: A bank is looking at a dataset of "Loan Applicants"

- One applicant has their "Income" and "Credit Score" listed, but their "Home Ownership" status (Owns vs. Rents) is missing
- Relationship: People with higher incomes and higher credit scores are statistically more likely to own their homes
- algorithm looks at thousands of other applicants to see the "probability" of owning a home based on income
- **Logistic regression:** Price $\rightarrow$ categorical data
- finds that this specific applicant (earning $120k with an 800 credit score) has a 92% probability of being a homeowner
- Result: probability $> 50\%$, algorithm imputes "Owns" into the missing cell

2. K-Nearest Neighbors (KNN) Imputation

- Identifies the K most similar records (neighbors) based on other available features and averages their values to fill the gap
  - similarity-based technique
- Strength: Excellent for capturing local patterns in the data
  - Very accurate for complex datasets
- Cons: Computationally expensive on large datasets

Example: Health Diagnostics

Imagine a hospital database tracking patient vitals. One patient (Patient X) has their Age and Weight recorded, but their Cholesterol Level is missing

Patient	Age	Weight	Cholesterol
A	50	85	240
B	22	60	180
C	48	82	230
D	25	62	190
X (Target)	49	84	NaN

Example: Health Diagnostics

1. Calculate Distance: similarity between Patient X and everyone else using the available data (Age and Weight).

- Patients A and C are very "close" to Patient X: similar age and weight
- Patients B and D are very "far": much younger and lighter

2. Pick "K" Neighbors: $K = 2$

- algorithm selects the two closest matches: Patient A and Patient C

3. Aggregate the Value: takes the average Cholesterol of those two neighbors

- Calculation: $(240 + 230)/2 = \mathbf{235}$

4. Impute: Patient X is assigned a Cholesterol value of 235

Why use KNN instead of Mean or Regression?

Better than Mean:

- Mean: Replaces with the average of the whole instances
  - impute everyone the same average value
    - A 22-year-old high-school grad and a 55-year-old PhD get the same income ($50k)
  - Data Relationships: Ignores correlations with other features
  - Variance: Reduces variance
  - Outliers: Sensitive to extreme outliers
- KNN: Uses the average of the most similar instances
  - far more realistic
    - 22-year-old HS grad gets income from other young HS grads $\approx 35K$
    - 55-year-old PhD gets income from other PhDs $\approx 120K$
  - Data Relationships: Preserves local correlations because neighbors are similar across multiple features
  - Variance: Preserves natural variance, maintains original data distribution
  - Outliers: Robust if k is chosen well

Why use KNN instead of Mean or Regression?

Better than Regression:

- Regression: global linear relationship
  - Fits one line: Price $= 50k + 150 \times \text{SqFt} + 20k \times \text{Bedrooms}$
    - fails: if price jumps non-linearly (e.g., luxury vs. economy markets) or if interactions matter
  - Overfitting: Can overfit if too many predictors are used
  - Outliers: Highly sensitive to outliers in predictors
- KNN: local similarity
  - Neighbors determine the value
  - Overfitting: Controlled by k (number of neighbors) - more robust

3. Multiple Imputation by Chained Equations (MICE)

- Instead of filling in a value once, MICE runs the imputation multiple times, creating several "possible" datasets
- analyzes all versions then averages the results
- accounts for the **uncertainty** of the missing data
  - leading to much better statistical integrity and more reliable confidence intervals
- Cons: Highly complex to implement and interpret

Duplicate Data

Duplicate Data

- same real-world entity is represented multiple times in a dataset
- common byproduct: merging different data sources (Data Integration) or manual entry errors
- If left unaddressed, duplicates can lead to overrepresentation, skewed statistical averages, and wasted computational resources

Deduplication Techniques

1. Exact Matching: looks for records that are 100% identical across specific key fields
   - Fields Used: Unique IDs (SSN, Passport Number), Email addresses, or a combination of Name + DOB
2. Fuzzy String Matching: Used when records are similar but not identical due to typos, nicknames, or different formatting
   - "Apple" and "Appel"
3. Entity Resolution: uses patterns across multiple fields to determine if two entries refer to the same person or object
   - Supervised ML: Training a model on a labeled dataset (pairs of "Match" vs "Non-match")
     - learns which differences matter (e.g., a middle name difference matters less than a birth year difference)
   - Unsupervised ML: Using clustering (like K-Means) to group similar records together
     - cluster is very tight, the records inside are likely duplicates

Inconsistent Data

Inconsistent Data

- Same values may appear with different spellings, cases, or formats
  - "Credit Card" vs "credit card"
  - Dates in different formats ($2023-01-05$, $01/07/2023$, $15-03-2023$)
  - "two" instead of 2 in quantity
- can lead to incorrect calculations and failed joins

Fixing Strategies

1. **Standardizing Units:** Data is mathematically incomparable until units are aligned
   - critical for financial and scientific datasets
   - Currencies: Converting all monetary values to a single base currency using a specific exchange rate
     - converting *EUR* and *GBP* to *USD*
   - Measurements: Converting weights, lengths, or temperatures
     - all temperature data is in *Celsius*

Fixing Strategies

2. **Harmonizing Formats:** Ensures that the structure of the data is identical across all rows, which is essential for sorting and filtering
   - Date Formats: Moving everything to the ISO 8601 standard (YYYY-MM-DD)
   - Capitalization: Ensuring all entries for a column like "City" follow the same casing
     - **LONDON**, **london**, and **London** → ***London***
3. **Resolving Naming Conflicts:** involves mapping different aliases or abbreviations to a single
   - e.g: **NY**, **N.Y.**, and **NYC** → ***New York***
   - Metadata Alignment: define that Cust_ID and Client_Num refer to the exact same attribute

Noise and Outlier Detection

Noise and Outlier Detection

- **Noise:** Random error or variance in a measured variable
  - Sensor glitch recording $-999^{\circ}C$
  - data entry typo "45" instead of "54"
- **Outliers:** Data points that differ significantly from the rest of the observations
  - deviations that may be caused by measurement errors
  - may be legitimate but rare occurrences
  - can skew models
    - 7-foot tall person
    - a transaction of $1M in a retail dataset

Approaches to Handle Noise: Smoothing

1. **Binning:** Data is first sorted and then distributed into "bins" (buckets)
   - Smooth by bin means: Replace all values in the bin with the average of that bin
   - Smooth by bin boundaries: Replace values with the closest boundary (minimum or maximum) of the bin
2. **Regression:** fitting the data to a mathematical function (like a straight line)
   - function captures the general pattern, and the random noise that deviates from the line is effectively ignored
3. **Moving Average Smoothing:** used in time-series data (like stock prices or weather)
   - calculates the average of a specific number of previous data points to create a "rolling" average
   - educes short-term fluctuations and highlights long-term cycles

Outlier Detection Methods

process of finding the **misfits** in data

1. Statistical Tests
   a. Z-Score: Measures how many standard deviations a point is from the mean
      - Z-score greater than 3 or less than -3 is flagged as an outlier
   b. Interquartile Range (IQR): Defines the middle 50% of the data
      - Anything that falls 1.5 times the IQR above the 3rd quartile or below the 1st quartile is considered an outlier
2. Clustering Techniques
   - DBSCAN (Density-Based Spatial Clustering of Applications with Noise) identifies outliers as points in low-density regions
     - far away from any crowd: automatically labeled as outliers
3. Distance-Based Methods
   - calculate the distance between every point and its neighbors
   - If a point's distance to its k-NN is much larger than the average distance in the dataset
     - isolated as an outlier

Data Integration

Data Integration

- process of combining data from multiple heterogeneous sources into a coherent, unified dataset
  - Hospital System Integration:
    - Emergency Room DB: patient_id = "ER-4521", name = "Bob Smith"
    - Pharmacy DB: cust_num = "P-8923", name = "Bob Smith"
    - Billing DB: account_no = "B-3341", name = "Bob Smith"
  - Without integration → Same patient appears as 3 different people
  - With integration → Unified patient view across all departments
- process is especially critical in big data environments where data silos are common
  - Data silos occur when data are separate and in non-communicating systems
- reduces redundancies and inconsistencies
  - significantly improving the accuracy and speed of the subsequent data mining process

Entity Identification Problem

Entity Identification Problem

- challenge of matching equivalent real-world entities from multiple data sources

1. Schema Integration: data from different sources have different schemas (structure, field names), aligning them becomes necessary
    a. Schema Matching: Identifying that customer_id in Database A is the same as cust_num in Database B
    b. Ontology-based mapping: Using a shared vocabulary or metadata standard to align schemas

Example

Source 1: CRM System

cust_id	name	phone
C1001	John Doe	555-1234
C1002	Jane Doe	555-5678

Source 2: ERP System

cust_num	full_name	ph_num
4521	J. Doe	555-1234
4522	J. Smith	555-9012

Observation:
C1001 (CRM) and 4521 (ERP) likely represent the same customer: same phone number, similar name

Example

- CRM Metadata for "status":
  - Name: status
  - Type: CHAR(1)
  - Values: 'A'(Active), 'I'(Inactive), 'P'(Pending)
  - Description: Customer account status
- ERP Metadata for “cust_status”:
  - Name: cust_status
  - Type: INTEGER
  - Values: 1(Active), 0(Inactive), 2(Pending)
  - Description: Customer status code
- Conclusion: These refer to the same entity but need value mapping
- Mapping: 'A'$\rightarrow$ 1, 'I'$\rightarrow$ 0, 'P'$\rightarrow$ 2

Redundancy and Correlation Analysis

Redundancy and Correlation Analysis

- dataset: redundancy if an attribute can be "derived" from another
- To detect redundancy, use Correlation Analysis:
  - **Nominal Data (Categorical):** use the χ^2 (Chi-square) test to see if two attributes are independent or related
  - **Numeric Data:** use Correlation Coefficients and Covariance to assess how values of one attribute vary relative to another

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- Special Case (Derived):
  - age = current_year - birth_year $\rightarrow$ Redundant
  - total_price = quantity $\times$ unit_price $\rightarrow$ Redundant
  - profit = revenue - cost $\rightarrow$ Redundant

Tuple Duplication

Tuple Duplication

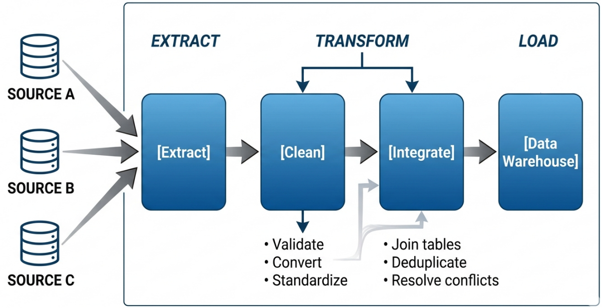
- Redundancy doesn't just happen in columns (attributes)
  - happens in rows (tuples)
- occurs when the same record exists multiple times in a dataset
- when the same unique data entry is represented by two or more identical (or nearly identical) rows
  - when merging tables that have been denormalized (flattened) for performance
- Denormalization: Instead of keeping "Customer Info" in one table and "Orders" in another
  - combine them into one giant table
    - causes the customer's name and address to be repeated for every single order they place
- Inconsistent Updates: If a customer moves houses and the address is updated on their new order
  - database now contains conflicting information for the same person

ETL & Data Warehousing

ETL & Data Warehousing

managed through an ETL pipeline:

- Extract: Getting data from sources like Apache Kafka (streaming) or SQL databases
- Transform: Solving the entity identification and conflict problems mentioned above
- Load: Moving data into a centralized Data Warehouse or Data Lake



Data Transformation

Data Transformation

- process of converting data into formats appropriate
  - optimal for analysis
- goal: more efficient and the resulting patterns easier to understand
- Strategies include:
  1. Normalization
  2. Discretization
  3. Compression
  4. Sampling

Normalization

Normalization

- involves scaling attribute data to fall within a small, specified range
  - 0.0 to 1.0 or -1.0 to 1.0
- Prevent attributes with larger ranges from dominating
  - Income ($0-200k) vs Age (0-80)

Dataset: Customer Records

Customer	Age	Income	Purchases
C1	25	50,000	12
C2	45	120,000	35
C3	30	75,000	18

Effect of Scale on Clustering Distance

Without Normalization:

- Age range: $25 - 45 \rightarrow$ range $= 20$
- Income range: $50,000 - 120,000 \rightarrow$ range $=$ 70,000
- Purchases range: $12\breve{~}35 \rightarrow$ range $= 23$

Distance between C1 and C2 (Euclidean):

$$\sqrt{(25-45)^2 + (50{,}000 - 120{,}000)^2 + (12-35)^2}$$

$$\approx \sqrt{400 + 4.9 \times 10^9 + 529} \approx 70{,}007$$

Observation: Income dominates the distance ($\approx 99.9\%$), while age and purchases contribute negligibly

Min-Max Normalization

- Performs a linear transformation on the original data

$$v_i' = \frac{v_i - \text{min}_A}{\text{max}_A - \text{min}_A}$$

- preserves the relationships among the original data values
- Sensitive to outliers
- Out-of-bounds error for new data outside original range
  - Requires storing min and max for future data
- Example: Normalize an income of \$73,600 into the range $[0.0, 1.0]$ where min $= \$12,000$ and max $= \$98,000$:

$$\frac{73,600 - 12,000}{98,000 - 12,000} = \mathbf{0.716}$$

Z-Score Normalization (Zero-Mean)

- Values are normalized based on the mean ($\bar{A}$) and standard deviation (σ_A)

$$v_i' = \frac{v_i - \bar{A}}{\sigma_A}$$

- Useful when the actual min/max are unknown or when outliers are present
- Example: If mean = \$54,000 and std dev = \$16,000, a value of \$73,600 becomes:

$$\frac{73,600 - 54,000}{16,000} = \mathbf{1.225}$$

Discretization

Discretization

- Converting continuous numeric attributes into discrete intervals or categorical labels
  - Raw Age: 25, 32, 18, 65, 42, 70, 22
  - Discretized (Equal-width, 3 bins):
    - 18-35: Youth; 36-52: Adult; 53-70: Senior
  - Result: Youth, Youth, Youth, Senior, Adult, Senior, Youth
- labels can form a concept hierarchy
  - Level 0: Price (raw values: 1, 5, 10, 15, 20, 25, 30, $\cdots$)
  - Level 1: (\$0$\cdots$\$100], (\$100$\cdots$\$200], (\$200$\cdots$\$300] (equal-width)
  - Level 2: (low, medium, high] (conceptual labels)
  - Level 3: (inexpensive, affordable, expensive] (semantic labels)
- useful when:
  - Building decision trees (which split on categorical conditions)
  - Reducing model complexity
  - Improving interpretability

Categorizations:

- Supervised: Uses class labels (target variable) while discretizing
- Unsupervised: Does not use class information (e.g., Binning, Histogram analysis)
- Top-down (Splitting): Starts with one interval and breaks it down
- Bottom-up (Merging): Starts with individual values and groups them

Discretization Techniques

A. Binning: unsupervised, top-down splitting technique

- Equal-width: Divides the range into N intervals of equal size
  - Sensitive to outliers

ages	[18, 22, 25, 30, 35, 40, 45, 50, 55, 60, 65, 70]
bins	4
Bin1	$[18.0, 31.0] \rightarrow 18, 22, 25, 30$
Bin2	$[31.0, 44.0] \rightarrow 35, 40$
Bin3	$[44.0, 57.0] \rightarrow 45, 50, 55$
Bin4	$[57.0, 70.0] \rightarrow 60, 65, 70$

- Equal-frequency (Equal-depth): Divides the range into N intervals
  - each containing approximately the same number of samples

salaries	[2500, 2800, 3000, 3200, 3500, 10000, 15000, 20000]
bins	2
Bin1	[2500, 3500] (4 values: 2500,2800,3000,3200)
Bin2	[3500, 20000] (4 values: 3500,10000,15000,20000)

Discretization Techniques

B. Histogram Analysis

- partitions continuous attribute values into disjoint ranges called buckets (or bins)
- Each bucket represents a range of values, and the height typically represents frequency (count) of values in that bucket

1. **Singleton Buckets:** Each bucket contains exactly one distinct value and its frequency count
2. **Equal-Width Histogram:** Each bucket covers the same numerical range (width [0-10]), regardless of how many data points fall into it
3. **Equal-Frequency (Equal-Depth) Histogram:** Each bucket contains (approximately) the same number of data points
    - Bucket boundaries vary to achieve equal counts